

# Scaling of the nonlinear response of the surface plasmon polariton at a metal/dielectric interface

Alexandre Baron,<sup>1,\*</sup> Stéphane Larouche,<sup>1</sup> Daniel J. Gauthier,<sup>2</sup> and David R. Smith<sup>1</sup>

<sup>1</sup>Center for Metamaterials and Integrated Plasmonics, Department of Electrical and Computer Engineering, Duke University, Durham, North Carolina 27708, USA

<sup>2</sup>Department of Physics, Duke University, Durham, North Carolina 27708, USA

\*Corresponding author: alexandre.baron@duke.edu

Received October 6, 2014; revised November 12, 2014; accepted November 12, 2014;  
posted November 14, 2014 (Doc. ID 223982); published December 4, 2014

Plasmonic systems involve interfaces containing metal and dielectric materials. In an effort to investigate the scaling of the nonlinear response of the surface plasmon polariton at a metal/dielectric interface, where the metal and dielectric present optical nonlinearity, we introduce a figure-of-merit that quantifies the contribution of the metal and the dielectric to the nonlinear response in this specific situation. In the case of self-action of the surface plasmon polariton for the gold/dielectric interface, we predict that the dielectric nonlinear response is dominant for strongly nonlinear dielectrics such as polydiacetylenes, chalcogenide glasses, or even semiconductors. The gold nonlinear response is dominant only in cases involving weakly nonlinear dielectrics such as silicon dioxide or aluminum oxide. We verify the relevance of the metric by investigating the process of optical switching via the third-order nonlinear response and discuss which gold/dielectric combinations have better switching behaviors. © 2014 Optical Society of America

OCIS codes: (240.6680) Surface plasmons; (190.0190) Nonlinear optics; (190.5940) Self-action effects.

<http://dx.doi.org/10.1364/JOSAB.32.000009>

## 1. INTRODUCTION

Nonlinear optics has enabled a variety of applications, including ultrafast lasers and amplifiers [1], optical frequency conversion [2], and nonlinear microscopy [3]. Another potential application that has garnered considerable interest in the past decades is all-optical switching [4]. Driven by the goal of making efficient nonlinear devices with low control power or high sensitivity, efforts have focused on studying nonlinear optics at the nanoscale, notably in plasmonic systems.

Surface plasmon polaritons (SPP) are oscillations of charge density waves coupled to photons at a metal–dielectric interface [5]. SPP stand out compared with other optical modes because their confinement is not restricted by the diffraction limit. As a result, an SPP concentrates fields at subwavelength scales to enhance the nonlinear response in metallic nanoparticles as well as on extended metal surfaces [6–8]. For almost two decades now, many theoretical proposals and experimental realizations have studied the nonlinear response of plasmonic systems, giving rise to the field of nonlinear plasmonics. Furthermore, the field of nonlinear metamaterials, in which artificial materials can be designed and fabricated that enhance nonlinear phenomena at optical frequencies, overlaps with nonlinear plasmonics [9].

Studies of nonlinear plasmonic systems or metamaterials include nonlinear plasmonic waveguides [10–15], nonlinear self-action [16], active control [17], nanofocusing [18], nonlinear switching [19,20], and four-wave mixing [21]. As was recently pointed out by several groups [22,23], most of these studies make the assumption that the response of the metal is linear and that the nonlinearity only originates from the dielectric medium adjacent to the metal. De Leon *et al.* [22] and Marini *et al.* [23] argue that past experimental work has shown

that the third-order nonlinear susceptibility  $\chi^{(3)}$  of metals can be enormous and can exhibit ultrafast behavior at optical wavelengths; this opens up the possibility of designing nonlinear plasmonic devices based on the intrinsic nonlinear response of the structure's metallic component.

There is a great deal of uncertainty regarding the value of  $\chi^{(3)}$  for metals (notably for gold) due to the strong frequency dispersion of metal nonlinear susceptibilities at wavelengths below 800 nm. Past measurements of  $\chi^{(3)}$  have exhibited a variation in magnitude of more than four orders of magnitude, with a strong dependence on pulse duration [24]. In spite of this, it is important to determine the scaling of the nonlinear response from the metal and dielectric for a range of experimental conditions such as laser wavelength, pulse duration, metal/dielectric combination, and geometry. For this reason, we introduce here an analytical figure-of-merit to assess the dominant nonlinear response of a nonlinear plasmonic system. We apply this metric to the simplest case of a plasmonic waveguide: a single metal/dielectric interface and focus on the case of gold to determine whether the dielectric or the gold nonlinear response dominates for several dielectrics. We study self-action of the SPP through the third-order nonlinearity and use full-wave numerical simulations to determine which nonlinear responses dominate in an optical switching scenario.

## 2. STRUCTURE INVESTIGATED AND DERIVATION OF THE FIGURE-OF-MERIT

In the following, we consider solely a metal/dielectric interface, as sketched in Fig. 1(a). This interface consists of a semi-infinite metal slab that occupies the region  $y < 0$ , which is adjacent to a dielectric that occupies the region  $y > 0$ . The

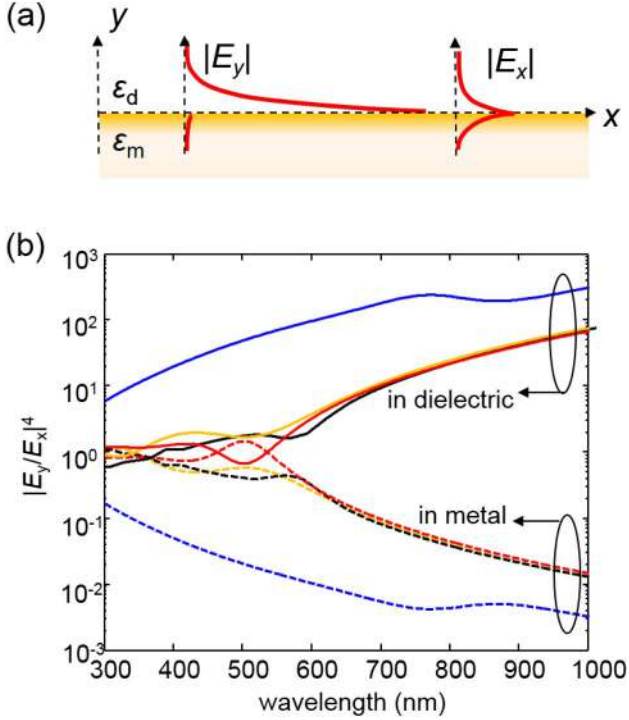


Fig. 1. Structure investigated and scaling of the electric field components. (a) Illustration of the single gold-dielectric interface. The transverse profiles of the modulus of both the  $y$  and  $x$  components of the field associated with the SPP mode are shown. (b) Spectral dependence of  $|E_y/E_x|^4$  in the dielectric (solid lines) and in the metal (dashed lines) for  $\epsilon_d = 5$  and for different metals (black; silver, red; gold, orange; copper and blue; aluminum).

transverse profiles of the  $y$  and  $x$  components of the electric field are sketched on the figure. The spatial dependence of the SSP field propagating in the positive  $x$  direction is given by  $\psi_{\text{SPP}}(x, y) = [H_z(x, y), E_x(x, y), E_y(x, y)]$  and may be written in the form [25]

$$\psi_{\text{SPP}}(x, y) = E_0 e^{ik_0 n_{\text{SPP}} x} \begin{cases} e^{ik_0 n_{\perp}^d y} \left[ 1/Z_0, -\frac{n_{\perp}^d}{\epsilon_d}, \frac{n_{\text{SPP}}}{\epsilon_d} \right], & \text{for } y > 0 \\ e^{-ik_0 n_{\perp}^m y} \left[ 1/Z_0, \frac{n_{\perp}^m}{\epsilon_m}, \frac{n_{\text{SPP}}}{\epsilon_m} \right], & \text{for } y < 0 \end{cases}, \quad (1)$$

where  $k_0 = 2\pi/\lambda_0$ ,  $n_{\text{SPP}} = \sqrt{\epsilon_d \epsilon_m / (\epsilon_d + \epsilon_m)}$  is the effective index of the SPP mode,  $n_{\perp}^d = \sqrt{\epsilon_d - n_{\text{SPP}}^2}$ ,  $n_{\perp}^m = \sqrt{\epsilon_m - n_{\text{SPP}}^2}$ ,  $Z_0$  is the impedance of free space,  $\epsilon_m$  is the dielectric constant of the metal, and  $\epsilon_d$  is that of the dielectric. From Eq. (1), we see that the ratio of the  $y$  component to the  $x$  component of the electric field in the dielectric (metal) is equal to  $|n_{\text{SPP}}/n_{\perp}^d|$  ( $|n_{\text{SPP}}/n_{\perp}^m|$ ). Because we are mostly interested in third-order nonlinear processes in isotropic materials in which the polarization mostly couples back to the SPP field (self-action), we find it useful to consider the ratio  $|E_y/E_x|^4$  in the metal and dielectric regions in order to determine the dominant field orientations and simplify the analytical expression for the figure-of-merit. Before focusing on the specific case of the gold interface with an arbitrary dielectric, we consider first the field component ratio for several metal interfaces having a common dielectric. For the spectral variations of  $\epsilon_m$ , we use data from Palik's handbook for gold and silver [26] and from

Rakic *et al.* [27] for aluminum and copper and obtain the results in Fig. 1(b), which compares the wavelength dependence of this ratio for  $\epsilon_d = 5$ , which is a typical value for chalcogenide glasses, for instance.

For all metals, the field component ratio is large in the dielectric and small in the gold for wavelengths toward the red end of the visible spectrum (i.e., above 600 nm), such that we may make the approximation that the total nonlinear polarization in the dielectric is mostly driven by the  $y$  component of the electric field, whereas the total nonlinear polarization in the metal is mostly driven by the  $x$  component of the electric field. This approximation is even more valid in the infrared, where the characteristic Drude model response of the metal is dominant, such that  $|E_y/E_x|$  scales more or less as  $\sqrt{\epsilon_m/\epsilon_d}$  in the dielectric and is vanishingly small in the metal region. It is interesting that this approximation is not valid at wavelengths below 600 nm where the interband transitions of metals dominate for gold, silver, and copper. On the other hand, aluminum, whose interband transition resides in the deep ultraviolet, displays identical behavior for all wavelengths above 300 nm.

Under these approximations and for the case of self-action, the third-order nonlinear polarization may be decoupled for each half-space as

$$P_d^{(3)}(x, y) = \frac{3}{4} \epsilon_0 \chi_d^{(3)} |E_y^d(x, y)|^2 E_y^d(x, y), \quad (2)$$

$$P_m^{(3)}(x, y) = \frac{3}{4} \epsilon_0 \chi_m^{(3)} |E_x^m(x, y)|^2 E_x^m(x, y), \quad (3)$$

where the  $d$  and  $m$  subscripts and superscripts refer to the dielectric and metal, respectively,  $P^{(3)}$  is the third-order nonlinear polarization, and  $\chi^{(3)}$  is the third-order nonlinear susceptibility. The relative importance of the dielectric nonlinear response in comparison with the metal nonlinear response is quantified by the ratio  $\rho$  of the spatial overlap integral in each medium between the nonlinear polarization and the SPP mode profile:

$$\rho = \left| \frac{\int P_d^{(3)}(x, y) E_d^*(x, y) dy}{\int P_m^{(3)}(x, y) E_m^*(x, y) dy} \right|. \quad (4)$$

This ratio is derived by treating the nonlinearity as a perturbation and considering its effect on the propagating SPP mode through coupled-mode theory [28,29]. A justification for considering this ratio is derived in Appendix A. Here, we only consider unidirectional propagation of the SPP mode and neglect any backscattering of the forward propagating mode and the nonlinear response. By introducing Eqs. (2) and (3) into Eq. (4) and making the approximation that  $E_d \sim E_y^d$  and  $E_m \sim E_x^m$  every time a fourth power of the modulus is involved, we obtain a convenient closed form expression for  $\rho$  given by

$$\rho = \left| \frac{\chi_d^{(3)}}{\chi_m^{(3)}} \left| \frac{\text{Im}(n_{\perp}^m)}{\text{Im}(n_{\perp}^d)} \right| \left| \frac{n_{\text{SPP}}}{\epsilon_d} \right|^4 \frac{\epsilon_m}{n_{\perp}^m} \right|. \quad (5)$$

This ratio is used below to determine which of the two nonlinear responses dominates, and it only depends on the material linear and nonlinear parameters.

### 3. SCALING OF THE NONLINEAR RESPONSE FOR DIFFERENT GOLD/DIELECTRIC COMBINATIONS

From Eq. (5), we see that  $\rho$  depends most importantly on the ratio of  $|\chi_m^{(3)}|/\chi_d^{(3)}$  and on  $\epsilon_d$ . We now focus on the case of a gold/dielectric interface. Figure 2 is a contour plot showing the scaling of  $\rho$  with these two quantities for an interface involving gold. The solid black curve represents the limiting case where  $\rho = 1$ , above which the gold nonlinear response is dominant, and below which the dielectric nonlinear response is dominant. This graph is convenient, as it enables us to identify the relevant nonlinear response for any gold/dielectric combination, solely based on the knowledge of  $\chi^{(3)}$  and  $\epsilon_d$ . The values used to compute Fig. 2 are given in Table 1. Of course, knowing the  $\chi^{(3)}$  of gold is paramount to computing this graph, and its value should be chosen carefully from the wide range of published values. Indeed, there is a strong dispersion of experimentally measured values of  $\chi^{(3)}$ , both with wavelength and with the pulse durations used to make the measurement, which is due to the many processes contributing to the third-order nonlinear response in gold [24]. Recently, De Leon *et al.* [30] measured a value of  $\chi^{(3)} = (4.67 + i3.03) \times 10^{-19} \text{ (m}^2/\text{V}^2\text{)}$  using the Kretschmann–Raether configuration at  $\lambda = 800 \text{ nm}$ . Their setup is remarkably sensitive to the nonlinear response of a thin gold film. The experiment was carried out with very short laser pulses ( $\sim 100 \text{ fs}$ ) for which the relevant nonlinearity is believed to be the smearing of the Fermi–Dirac distribution function due to hot electrons [31]. As the pulse duration is increased, the third-order nonlinearity gradually transfers to a thermal nonlinearity with heating of the lattice due to the hot electron relaxation, yielding values that are typically 100 times larger than the ultrafast value and which are often used to make theoretical predictions [22,23].

To produce Table 1 and compute  $\rho$ , we used the value measured by De Leon *et al.* [30]. Figure 2 shows that the nonlinear

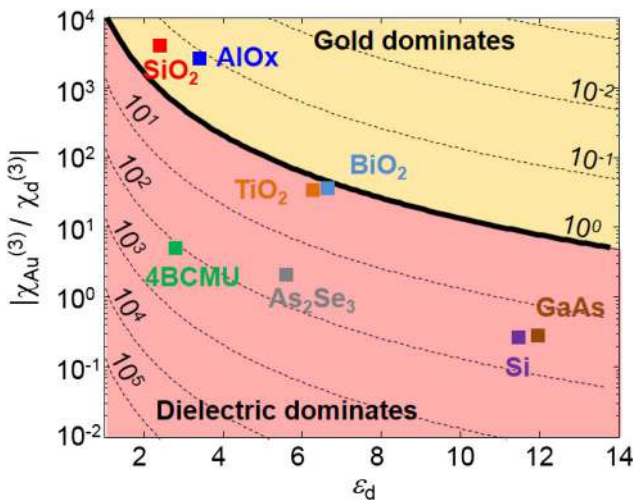


Fig. 2. Mapping of the nonlinear response for a gold/dielectric interface. Contour plot of  $\rho$ , given by Eq. (5) as a function of the modulus of the ratio of the third-order nonlinear susceptibilities of gold and the dielectric material and of the real part of  $\epsilon_d$ . The black continuous curve denotes the  $\rho = 1$  limiting case above which the gold nonlinear response is stronger and below which the dielectric response is stronger. Several nonlinear dielectrics are indicated on the graph.

response of gold dominates for the case of weakly nonlinear dielectrics such as silicon dioxide and aluminum oxide. However, for strongly nonlinear dielectrics such as highly nonlinear polymers [polydiacetylenes (4BCMU), chalcogenide glasses ( $\text{As}_2\text{Se}_3$ ), or semiconductors (GaAs and Si)], it is the dielectric nonlinear response that is dominant. Some cases are more difficult to determine, such as with titanium oxide or bismuth oxide and are likely to involve competition between the nonlinearities in each domain. It should be noted that the graph was produced for  $\lambda = 800 \text{ nm}$ , but certainly captures the essential physics for wavelengths in the near-infrared since  $\chi^{(3)}$  of gold is mostly constant for wavelengths greater than  $750 \text{ nm}$  [23], and since the SPP mode profile remains almost identical because of weak dispersion for most of the dielectrics considered here. For shorter wavelengths, obviously, the graph will be greatly modified due to the strong wavelength dispersion of  $\chi^{(3)}$  of gold and the SPP.

### 4. FULL-WAVE SIMULATIONS: THE CASE OF OPTICAL SWITCHING OF THE SPP

Next, we use full-wave numerical simulations using the commercial software COMSOL Multiphysics based on the finite-element method and verify that the behavior shown in Fig. 2 can be used to understand optical switching of the SPP for a subset of dielectrics.

Optical switching of the SPP is a self-induced process in which the third-order nonlinear polarization given by Eqs. (2) and (3) produces an intensity-dependent modulation of the dielectric constant, given by

$$\epsilon = \epsilon_L + \frac{3}{4}\chi^{(3)}|E|^2, \quad (6)$$

where  $\epsilon_L$  is the linear dielectric constant. This modulation, in turn, produces a modulation of the SPP effective mode index  $n_{\text{SPP}}$ , which causes a decrease in the SPP propagation length, defined as the  $1/e$  decay length of the SPP intensity. In the linear case, it is simply equal to

$$l_{\text{SPP}} = \frac{\lambda}{4\pi \text{Im}[n_{\text{SPP}}]}. \quad (7)$$

Table 1. Linear and Nonlinear Parameters of Dielectrics Considered in Our Simulations<sup>a</sup>

| Material                 | $\epsilon_d$ | $\chi_d^{(3)} \text{ (m}^2/\text{V}^2\text{)} \times 10^{-19}$ | $ \chi_{\text{Au}}^{(3)} /\chi_d^{(3)}$ | $\lambda \text{ (}\mu\text{m)}$ | Reference |
|--------------------------|--------------|----------------------------------------------------------------|-----------------------------------------|---------------------------------|-----------|
| $\text{Al}_2\text{O}_3$  | 3.24         | 0.0031                                                         | 1800                                    | 1.06                            | [32]      |
| GaAs                     | 12           | $24 + i48$                                                     | 0.2                                     | 1.55                            | [33]      |
| Si                       | 11.6         | $25 + i0.0039$                                                 | 0.2                                     | 1.55                            | [34]      |
| $\text{TiO}_2$           | 6.15         | 0.21                                                           | 27                                      | 1.06                            | [32]      |
| $\text{SiO}_2$           | 2.25         | 0.0025                                                         | 2200                                    | 1.3                             | [32]      |
| $\text{As}_2\text{Se}_3$ | 5.76         | $4.1 + i0.56$                                                  | 1.4                                     | 1.5                             | [35]      |
| $\text{Bi}_2\text{O}_3$  | 6.25         | 0.24                                                           | 23                                      | 1.5                             | [35]      |
| 4BCMU                    | 2.43         | $-1.3 + i0.55$                                                 | 4.3                                     | 1.06                            | [32]      |
| Air                      | 1.0006       | $1.7 \times 10^{-6}$                                           | $3 \times 10^6$                         |                                 | [32]      |

<sup>a</sup>Values have been gathered either from direct values of  $\chi^{(3)}$  or from values of the intensity-dependent refractive index  $n_2 = 3 \text{Re}[\chi^{(3)}]/(4n_0^2\epsilon_0 c)$  (when  $n_2$  has units of  $\text{m}^2/\text{W}$ ) and the nonlinear absorption coefficient  $\beta = 3\omega \text{Im}[\chi^{(3)}]/(2n_0^2\epsilon_0 c^2)$ , when relevant. The value used here for the  $\chi^{(3)}$  of gold is  $(4.67 + i3.03) \times 10^{-19} \text{ (m}^2/\text{V}^2\text{)}$  and is taken from De Leon *et al.* [30].



In some cases, the modulation of  $n_{\text{SPP}}$  can produce an increase of  $l_{\text{SPP}}$ . The relative increase or decrease in  $l_{\text{SPP}}$  depends on the signs of the real and imaginary parts of  $\chi^{(3)}$ . For instance, we expect an increase of  $l_{\text{SPP}}$  at  $\lambda = 500$  nm, where the real and imaginary parts of  $\chi^{(3)}$  of gold are negative.

The total electromagnetic field of the single interface problem is computed in a 2D configuration when excited by the SPP mode for a fixed waveguide length. The semi-infinite half-planes (gold and dielectric) are defined with a diagonal dielectric constant tensor:

$$\varepsilon = \varepsilon_L I_3 + \frac{3}{4} \chi^{(3)} \begin{bmatrix} |E_x|^2 & 0 & 0 \\ 0 & |E_y|^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad (8)$$

where  $I_3$  is the  $3 \times 3$  identity matrix and where the relevant values of  $\varepsilon_L$  and  $\chi^{(3)}$  are introduced in each medium. The calculations are carried out as the input power per unit length  $S_0$  carried by the SPP is varied. The power per unit length  $S(x)$  along the  $z$  direction is defined as the power flow along the  $x$  direction of the time-averaged Poynting vector  $S(x) = 1/2 \text{Re} \int_{-\infty}^{+\infty} (\mathbf{E} \times \mathbf{H}) \cdot \mathbf{e}_x dy$ . Figure 3 shows the field distribution in the case of a gold/GaAs interface for different values of  $S_0$  over a length equal to the SPP propagation length in the linear case. As expected, we observe that the field extends over smaller and smaller portions of the waveguide as  $S_0$  is increased. This is due to the self-induced modulation of the effective mode index.

Next, we numerically calculate the propagation length of the SPP for several gold/dielectric combinations to determine its dependence with  $S_0$  (see Fig. 4). For every combination,  $S_0$  is varied to quantify the transition from the linear to the nonlinear regime; calculations are made by considering the nonlinear response of gold only (crosses) and of the gold and dielectric (circles). On the one hand, the threshold values of  $S_0$ , where the switching behavior occurs, are lower by about two orders of magnitude for the cases of GaAs and 4BCMU when the dielectric nonlinear response is included, clearly demonstrating that the nonlinearity of the dielectric dominates. The cases of air and silica, on the other hand, are very different, as there is no noticeable difference in the optical switching of the SPP, whether or not the

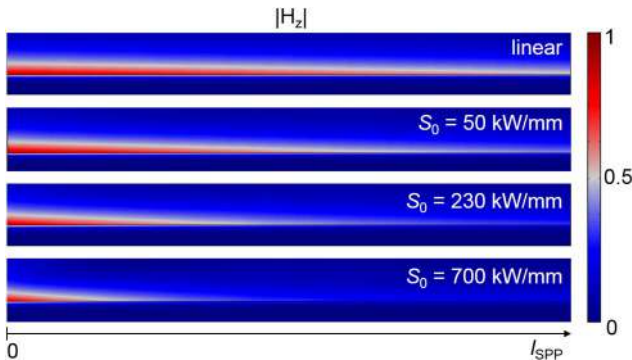


Fig. 3. Nonlinear propagation of the SPP field. Modulus of the magnetic field  $H_z$  calculated for a gold/GaAs interface at  $\lambda = 1.5 \mu\text{m}$  over a length equal to the propagation length of the surface plasmon in the linear regime for different values of  $S_0$ . The linear and nonlinear properties of the GaAs are given in Table 1. Each plot is normalized to the input value of  $|H_z|$ .

nonlinearity of the dielectric is included, which shows that the gold nonlinear response dominates. Both of these observations are expected from the mapping of  $\rho$  (see Fig. 2).

The representation provided in Fig. 4 is quite convenient, as it can be used to determine the potential size of an SPP switching device and the minimal control power required for operation. Bearing this in mind, we can provide a discussion on the values of  $S_0$  in these simulations. First, it should be noted that the switching thresholds of  $S_0$  scale proportionally with the inverse of  $\chi^{(3)}$ . This is indicated by the dotted curve in Fig. 4, which corresponds to a simulation of the gold/air interface when the  $\chi^{(3)}$  of gold is equal to that found in [22,23], which is about two orders of magnitude larger than the  $\chi^{(3)}$  used for the remaining simulations. The variation of the propagation length in this case occurs at values of  $S_0$  that are within an order of magnitude of those predicted by the De Leon *et al.* model [22], which investigates small variations of the propagation length ( $\sim 10\%$ ). Second, the dashed line in Fig. 4 indicates the damage threshold of gold for pulses on the order of 100 fs [36], which shows that, in the specific situation studied here, the single gold/air and gold/SiO<sub>2</sub> interfaces would actually constitute poor optical switches given the  $\chi^{(3)}$  for gold used in our present calculations. Higher  $\chi^{(3)}$  values will reduce the switching threshold. So, third-order processes based on the slower thermal nonlinearities of gold, which produces higher effective  $\chi^{(3)}$  (i.e., in the pico and nanosecond regime), will make better switches in the cases where air or SiO<sub>2</sub> is used as the dielectric. Some dielectrics (4BCMU and GaAs) enable operation well below the optical damage threshold. As a result, though the propagation length of the SPP is shorter when using high dielectric constant materials, they constitute smaller switches with lower command powers compared with switches based on the intrinsic nonlinearity of gold. For the value of  $\chi^{(3)}$  used here, making  $\rho$  as large

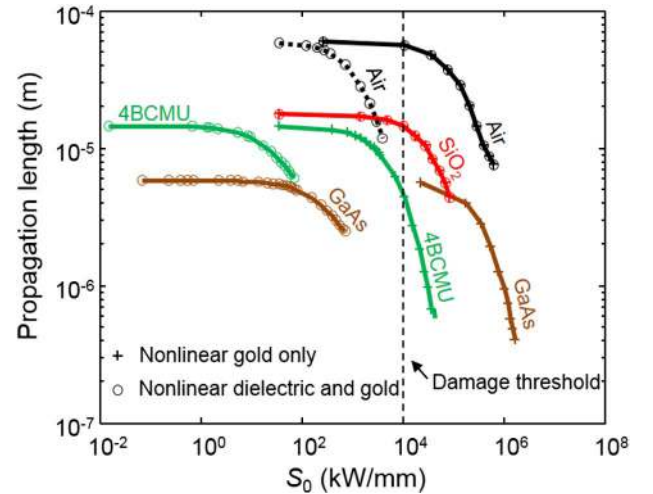


Fig. 4. Optical switching of the SPP for different gold/dielectric combinations. Propagation length of the SPP as a function of the input power per unit length  $S_0$  for different dielectrics (black, air; red, SiO<sub>2</sub>; green, 4BCMU; brown, GaAs). Crosses correspond to calculations that are carried out when only the gold nonlinear response is taken into account. Rings correspond to calculations that are carried out when the gold and dielectric nonlinear response is taken into account. The dotted curve is computed for the gold/air interface with the value of  $\chi^{(3)}$  taken from Marini *et al.* [19]. The dashed line indicates the optical damage threshold of gold for 100 fs pulses.

as possible optimizes the switch. We stress that our simulations represent a best case scenario in that we do not consider competing nonlinearities that are likely to appear when  $S_0$  becomes large, such as free-carrier nonlinearities in GaAs or Si [33,34]. However, the metric we have introduced and the analysis we provide can be applied to other situations involving other nonlinear processes and different waveguide geometries to guide (1) the choice of the dielectric adjacent to gold; (2) the device length; and (3) the command power required to operate a plasmonic device for a specific nonlinear application.

In summary, we introduce a figure-of-merit  $\rho$ , which enables us to predict the dominant nonlinear response (metal or dielectric) for a given plasmonic problem. We evaluate this metric for the case of a single gold/dielectric interface for several nonlinear dielectrics and investigate configurations where the gold nonlinear response dominates and others where the dielectric nonlinear response dominates. Finally, we focus on self-action of the SPP through the third-order nonlinearity and verify which of the nonlinear responses dominates in the case of optical switching. Since our approach describes the nonlinear polarization in terms of the local SPP field and focuses on how it will act on the SPP mode itself in the metal and in the dielectric, our metric fully accounts for the plasmonic enhancement of nonlinear effects in the dielectric. Additional work is required to capture the essential physics that govern the complex nonlinear response of simple plasmonic architectures. The metric we have introduced can be applied to finite thickness metallic slabs surrounded by dielectrics or to finite thickness dielectric slabs surrounded by metals. Both structures present symmetric and antisymmetric modes, which are likely to present very different nonlinear behaviors that depend on the slab thickness because the transverse profiles of the electric field components are drastically different. In such cases, however,  $\rho$  cannot be determined analytically because the dispersion relations of these waveguides do not have an analytical solution.

## APPENDIX A

The single metal/dielectric interface is a single-mode waveguide for which the SPP field  $\mathbf{E}_{\text{SPP}}$  is a solution of the propagation equation

$$(\nabla^2 - \beta_{\text{SPP}}^2)\mathbf{E} = 0, \quad (\text{A1})$$

where  $\beta_{\text{SPP}} = k_0 n_{\text{SPP}}$  is the SPP propagation constant. Considering the third-order nonlinear polarization presented in Eqs. (2) and (3) to be a small perturbation, the nonlinear propagation equation of the perturbed case is given by

$$(\nabla^2 - \beta_{\text{SPP}}^2)\mathbf{E} = \mu_0 \omega^2 \mathbf{P}^{(3)}. \quad (\text{A2})$$

From perturbation theory, we may consider  $\alpha \mathbf{E}_{\text{SPP}}$  to be a solution to Eq. (A2), with  $|\alpha| \sim 1$  and  $\arg(\alpha) \sim 0$ . Then, by making the slowly varying envelope approximation ( $\|\mathbf{E}_{\text{SPP}} \nabla^2 \alpha\| \ll \|2\nabla \alpha \nabla \mathbf{E}_{\text{SPP}}\|$ ) and using Eq. (A1), Eq. (A2) reduces to

$$\mathbf{E}_{\text{SPP}} \nabla \alpha = -\frac{i\mu_0 \omega^2}{2\beta_{\text{SPP}}} \mathbf{P}^{(3)}. \quad (\text{A3})$$

Applying the dot product with the complex conjugate of  $\mathbf{E}_{\text{SPP}}$  to both sides of Eq. (A3) and integrating over the entire volume  $\Omega$  of space, we obtain the relation

$$\nabla \alpha = -\frac{i\mu_0 \omega^2}{2\beta_{\text{SPP}} \Omega_{\text{SPP}}} \left( \int_{\Omega_m} \mathbf{P}^{(3)} \mathbf{E}_{\text{SPP}}^* d\Omega + \int_{\Omega_d} \mathbf{P}^{(3)} \mathbf{E}_{\text{SPP}}^* d\Omega \right), \quad (\text{A4})$$

where  $\Omega_m$  and  $\Omega_d$  represent the metal and dielectric half-volumes, respectively, and  $\Omega_{\text{SPP}} = \int_{\Omega} |\mathbf{E}_{\text{SPP}}|^2 d\Omega$  is the SPP mode volume. Factoring out the integral over the dielectric half-volume in the previous equation yields

$$\nabla \alpha = -\frac{i\mu_0 \omega^2}{2\beta_{\text{SPP}} \Omega_{\text{SPP}}} \int_{\Omega_m} \mathbf{P}^{(3)} \mathbf{E}_{\text{SPP}}^* d\Omega \left( 1 + \frac{\int_{\Omega_d} \mathbf{P}^{(3)} \mathbf{E}_{\text{SPP}}^* d\Omega}{\int_{\Omega_m} \mathbf{P}^{(3)} \mathbf{E}_{\text{SPP}}^* d\Omega} \right). \quad (\text{A5})$$

We define the real quantity  $\rho$  as the magnitude of the ratio in Eq. (A5):

$$\rho = \left| \frac{\int_{\Omega_d} \mathbf{P}^{(3)} \mathbf{E}_{\text{SPP}}^* d\Omega}{\int_{\Omega_m} \mathbf{P}^{(3)} \mathbf{E}_{\text{SPP}}^* d\Omega} \right|. \quad (\text{A6})$$

It is clear that  $\rho \ll 1$  ( $\rho \gg 1$ ) implies that the nonlinear response is dominated by the metal (dielectric) half-volume. Because the single metal/dielectric interface is a 1D problem, the volume integrals in Eq. (A6) reduce to line integrals along a line orthogonal to the propagation direction, which yields Eq. (4). The calculation made here implicitly assumes that the nonlinear polarization included in Eq. (A2) shares the same frequency as the electric field of the SPP mode.

It should be noted that, in the general case of subwavelength waveguides, such as microstructured fibers, a full vectorial model of nonlinear propagation is required to account for the role of the longitudinal component of the electric field, which can result in a totally different nonlinear behavior to that expected from classical nonlinear models [37]. The reason for this is that the fundamental mode can have two polarizations, which are not orthogonal to one another in the classical sense (i.e., for the transverse fields), if the longitudinal component of the electric field is strong. In such cases, the nonlinear propagation equation takes a different form to that presented in Eq. (A4). We refer readers to Afshar *et al.* [37] for further reading on the subject, but this observation does not apply to plasmonic waveguides because the fundamental mode only has a single polarization that is transverse magnetic.

## ACKNOWLEDGMENTS

A. B. would like to thank Chistos Argyropoulos and Britt Lassiter for useful discussions on plasmonics and Patrick Bowen for discussions on coupled mode theory.

## REFERENCES

1. G. Cerullo and S. De Silvestri, "Ultrafast optical amplifiers," *Rev. Sci. Instrum.* **74**, 1–18 (2003).
2. M. Fejer, "Nonlinear optical frequency conversion," *Phys. Today* **47**, 25–32 (1994).
3. Y. Wang, C.-Y. Lin, A. Nikolaenko, V. Raghunathan, and E. O. Potma, "Four-wave mixing microscopy of nanostructures," *Adv. Opt. Photon.* **3**, 1–52 (2011).

4. D. Cotter, R. Manning, K. Blow, A. Ellis, and A. Kelly, "Nonlinear optics for high-speed digital information processing," *Science* **286**, 1523–1528 (1999).
5. H. Raether, *Surface Plasmons on Smooth and Rough Surfaces and on Gratings* (Springer, 1998).
6. J. A. Schuller, E. S. Barnard, W. Cai, Y. C. Jun, J. S. White, and M. L. Brongersma, "Plasmonics for extreme light concentration and manipulation," *Nat. Mater.* **9**, 193–204 (2010).
7. J. Renger, R. Quidant, N. van Hulst, and L. Novotny, "Surface-enhanced nonlinear four-wave mixing," *Phys. Rev. Lett.* **104**, 059903 (2010).
8. A. Bouhelier, M. Beversluis, A. Hartschuh, and L. Novotny, "Near-field second-harmonic generation by local field enhancement," *Phys. Rev. Lett.* **90**, 013903 (2003).
9. N. I. Zheludev, "The road ahead for metamaterials," *Science* **328**, 582–583 (2010).
10. D. Mihalache, G. I. Stegeman, C. T. Seaton, E. M. Wright, R. Zanon, A. D. Boardman, and T. Twardowski, "Exact dispersion relations for transverse magnetic polarized guided waves at a nonlinear interface," *Opt. Lett.* **12**, 187–189 (1987).
11. H. Yin, C. Xu, and P. M. Hui, "Exact surface plasmon dispersion relations in a linear-metal-nonlinear dielectric structure of arbitrary nonlinearity," *Appl. Phys. Lett.* **94**, 221102 (2009).
12. J.-H. Huang, R. Chang, P.-T. Leung, and D. P. Tsai, "Nonlinear dispersion relation for surface plasmon at a metal-Kerr medium interface," *Opt. Commun.* **282**, 1412–1415 (2009).
13. A. R. Davoyan, I. V. Shadrivov, and Y. S. Kivshar, "Nonlinear plasmonic slot waveguides," *Opt. Express* **16**, 21209–21214 (2008).
14. A. Degiron and D. R. Smith, "Nonlinear long-range plasmonic waveguides," *Phys. Rev. A* **82**, 033812 (2010).
15. Y. Liu, G. Bartal, D. A. Genov, and X. Zhang, "Subwavelength discrete solitons in nonlinear metamaterials," *Phys. Rev. Lett.* **99**, 153901 (2007).
16. E. Feigenbaum and M. Orenstein, "Plasmon-soliton," *Opt. Lett.* **32**, 674–676 (2007).
17. K. F. MacDonald, Z. L. Samson, M. I. Stockman, and N. I. Zheludev, "Ultrafast active plasmonics," *Nat. Photonics* **3**, 55–58 (2008).
18. A. R. Davoyan, I. V. Shadrivov, A. A. Zharov, D. K. Gramotnev, and Y. S. Kivshar, "Nonlinear Nanofocusing in Tapered Plasmonic Waveguides," *Phys. Rev. Lett.* **105**, 116804 (2010).
19. R. E. Noskov, P. A. Belov, and Y. S. Kivshar, "Subwavelength modulational instability and plasmon oscillons in nanoparticle arrays," *Phys. Rev. Lett.* **108**, 093901 (2012).
20. R. E. Noskov, A. E. Krasnok, and Y. S. Kivshar, "Nonlinear metal-dielectric nanoantennas for light switching and routing," *New J. Phys.* **14**, 093005 (2012).
21. H. Suchowski, K. O'Brien, Z. J. Wong, A. Salandrino, X. Yin, and X. Zhang, "Phase-mismatch-free nonlinear propagation in optical zero-index materials," *Science* **342**, 1223–1226 (2013).
22. I. De Leon, J. E. Sipe, and R. W. Boyd, "Self-phase modulation of surface plasmon polaritons," *Phys. Rev. A* **89**, 013855 (2014).
23. A. Marini, M. Conforti, G. Della Valle, H. W. Lee, Tr. X. Tran, W. Chang, M. A. Schmid, S. Longhi, P. St. J. Russell, and F. Biancalana, "Ultrafast nonlinear dynamics of surface plasmon polaritons in gold nanowires due to the intrinsic nonlinearity of metals," *New J. Phys.* **15**, 013033 (2013).
24. R. W. Boyd, Z. Shi, and I. De Leon, "The third-order nonlinear optical susceptibility of gold," *Opt. Commun.* **326**, 74–79 (2014).
25. P. Lalanne, J. P. Hugonin, H. T. Liu, and B. Wang, "Theory of surface plasmon generation at nanoslit apertures," *Surf. Sci. Rep.* **64**, 453–469 (2009).
26. E. D. Palik, *Handbook of Optical Constants of Solids* (Academic, 1985).
27. A. D. Rakic, A. B. Djuricic, J. M. Elazar, and M. L. Majewski, "Optical properties of metallic films for vertical-cavity optoelectronic devices," *Appl. Opt.* **37**, 5271–5283 (1998).
28. A. Yariv, "Coupled-mode theory for guided-wave optics," *IEEE J. Quantum Electron.* **9**, 919–933 (1973).
29. G. P. Agrawal, *Nonlinear Fiber Optics*, 4th ed. (Academic, 2007), Chap. 2.
30. I. De Leon, Z. Shi, A. C. Liapis, and R. W. Boyd, "Measurement of the complex nonlinear optical response of a surface plasmon-polariton," *Opt. Lett.* **39**, 2274–2277 (2014).
31. F. Hache, D. Ricard, C. Flytzanis, and U. Kreibig, "The optical Kerr effect in small metal particles and metal colloids: the case of gold," *Appl. Phys. A* **47**, 347–357 (1988).
32. R. W. Boyd, *Nonlinear Optics*, 3rd ed. (Academic, 2008), Chap. 4.
33. A. Baron, A. Rysanyanskiy, N. Durbreuil, P. Delaye, Q. V. Tran, S. Combrié, A. de Rossi, R. Frey, and G. Roosen, "Light localization induced enhancement of third order nonlinearities in a GaAs photonic crystal waveguide," *Opt. Express* **17**, 552–557 (2009).
34. A. Baron, N. Durbreuil, P. Delaye, R. Frey, and G. P. Agrawal, "Raman amplification of optical pulses in silicon nanowaveguides: impact of spectral broadening of pump pulses," *J. Eur. Opt. Soc.* **6**, 11030 (2011).
35. B. J. Eggleton, B. Luther-Davies, and K. Richardson, "Chalcogenide photonics," *Nat. Photonics* **5**, 141–148 (2011).
36. J. Krüger, D. Dufft, R. Koter, and A. Hertwig, "Femtosecond laser-induced damage of gold films," *Appl. Surf. Sci.* **253**, 7815–7819 (2007).
37. V. S. Afshar and T. M. Monro, "A full vectorial model for pulse propagation in emerging waveguides with subwavelength structures part I: Kerr nonlinearity," *Opt. Express* **17**, 2298–2318 (2009).